

**Cardinal Santos Medical Center
Department of Internal Medicine**

**Factors affecting mental health outcomes of healthcare workers during
Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center**

In Partial Fulfillment of the Requirements for the
Internal Medicine Residency Training Program

Archimedes A. Apa-ap, MD
Rosa Allyn G. Sy, MD, FPCP, FPSEDM
Marc Conrad C. Molina, MD, FPNA

October 2020

Table of Contents

Abstract	3
1. Introduction	4
Background of the study	4
Research Question	5
Objective of the study	5
Significance of the study	6
Conceptual Framework	6
Scope and Limitations	7
Definition of Terms	7
2. Methodology	9
Research Design	9
Statistical Analysis	12
Ethical Consideration	12
3. Results	12
4. Discussion	27
5. Conclusion and Recommendation	28
Appendix	29
References	32

Abstract

Background. The outbreak of the novel Coronavirus Disease (COVID-19) was declared a pandemic by the World Health Organization on March 2020. Studies show that healthcare workers from previous pandemics (A/H1N1 influenza; severe acute respiratory syndrome, SARS), have experienced high levels of stress, anxiety and low mood with negative psychological impacts sustained after one year. This paper aims to identify factors affecting mental health outcomes of healthcare workers during Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center.

Methods. This is a cross-sectional study of factors affecting mental health outcome of healthcare workers with direct contact to a patient with confirmed or suspected case Coronavirus Disease (COVID-19) at Cardinal Santos Medical Center from March 2020 to September 2020. The severity of mental health outcome in terms of depression, anxiety and distress were measured using the 9-item Patient Health Questionnaire, the 7-item Generalized Anxiety Disorder scale, and the 22-item Impact of Event Scale–Revised, respectively. Binomial multiple logistic regression model was used to identify the factors associated with mental health outcome.

Results. A total of 172 healthcare workers were included in this study. Prevalence of depression, anxiety and distress symptoms among all healthcare workers were 57%, 47.7%, 65.1%, respectively. After multivariate analysis, age and living with a family with comorbidities are significantly associated with depression and anxiety. Healthcare workers aged 31 years and above are 70% less likely to have depression symptoms (OR 0.278, CI 95%, 0.11-0.72 $p=0.008$ and OR 0.273, CI 95%, 0.12-0.61, $p=0.002$). Those living with relatives with comorbidities are 2.7 times more likely to experience depression symptoms (OR 2.731, CI 95%, 1.36-5.5, $p=0.005$). Healthcare workers age 31-40 years has 80% less likely to experience anxiety symptoms (OR 0.212, CI 95%, 0.09-0.5, $p<0.001$), and those age 41 and above have 73% less likely to have anxiety symptoms (OR 0.37, CI 95%, 0.14-0.98, $p<0.045$). Those living with relatives with comorbidities are 2.9 times more likely to have anxiety symptoms (OR 2.889, CI 95%, 1.46-5.73, $p=0.002$). Level of knowledge on COVID-19 among health care workers significantly differs across severity but has no significant association with prevalence of depression. No factors were associated with distress symptom.

Conclusion. Emotional strain by health care workers during pandemic can cause adverse outcomes not only for individual wellbeing, but also for patient care and the healthcare system. Cognizant of high rates of depression, anxiety and distress among our younger health care workers living with relatives with comorbidities during this time will be a challenge to both hospital management and the government to establish and/or improve existing programs to promote mental wellbeing and address their mental health needs. These programs should be readily available and accessible to healthcare workers seeking psychological support. Psychological stress can accumulate over time and have an impact later in the outbreak, including posttraumatic stress disorder (PTSD), which should be investigated in future studies.

1. Introduction

Background of the study

The novel Coronavirus disease (COVID-19) was first reported in December 2019 in Wuhan, China. ¹ The World Health Organization (WHO) in January 2020, declared the outbreak of the novel coronavirus disease (COVID-19) to be a Public Health Emergency of International Concern. Coronavirus Disease (COVID-19) was then declared to be a pandemic by March 2020. ² As of October 5, 2020, there are over 34.8 million recorded cases and over 1 million recorded deaths of Coronavirus Disease (COVID-19) globally by the World Health Organization. ³ The Philippines' Department of Health confirmed 329,637 cases and 5,925 of deaths from Coronavirus Disease (COVID-19) as of October 7, 2020. ⁴ The number of health workers who tested positive for the coronavirus was 5,008, with majority of infections seen in nurses and physicians. The top 5 medical professions that reported coronavirus cases are nurses (1,734 cases), physicians (1,100), nursing assistants (338), medical technologists (210), and radiologic technologists (119) as of August 3, 2020. ⁵

Studies show that healthcare workers from previous pandemics (A/H1N1 influenza; severe acute respiratory syndrome, SARS), have experienced high levels of stress, anxiety and low mood with negative psychological impacts sustained after one year ⁶⁻⁷⁻⁸ A number of factors including increasing number of confirmed and suspected cases, Irregular working hours and overwhelming workload, depletion of personal protection equipment, widespread media coverage, lack of specific drugs, and feelings of being inadequately supported may all contribute to the mental burden of these healthcare workers. ⁹⁻¹⁰ The extreme pressures experienced by healthcare workers during a pandemic may increase their risk of burnout, which has adverse outcomes not only for individual wellbeing, but also for patient care and the healthcare system. ¹¹ Lai, J. et. al. reported that health workers directly engaged in the care of patients with Coronavirus Disease (COVID-19) experienced symptoms of depression, anxiety, and distress by 50.4%, 44.6% and 71.5% respectively. Most were female health workers, aged 26-40 years old, married, with a junior technical title. ¹² A similar research done by Kang, L. Et. Al revealed 36% percent of the medical staff had subthreshold mental health disturbances, 34.4% had mild disturbances, 22.4% had moderate disturbances and 6.2% had severe disturbances, in the immediate wake of the viral epidemic. Most were female (85.5%), aged 25 to 40 years (63.4%), married (56.9%), have an educational level of undergraduate or less (85%), and have a junior technical title (66.3%). 17 to 50% had had participated in counseling or psychotherapy, had accessed psychological materials, and psychological resources available through media. Trends in levels of psychological distress and factors such as exposure to infected people and psychological assistance were identified. These findings emphasize the importance of being prepared to support frontline workers through mental health interventions at times of widespread crisis. ¹³ These shows that concerns about the mental health, psychological adjustment, and recovery of healthcare workers treating and caring for patients with Coronavirus Disease (COVID-19) are now arising. ¹²⁻¹³

Knowledge of a disease may influence a healthcare workers' attitudes and practices, and incorrect attitudes and practices directly increase the risk of infection.¹⁴ Older people and people with pre-existing medical conditions (such as diabetes, heart disease, preexisting heart condition, chronic obstructive lung disease, asthma) appear to be more vulnerable to becoming severely ill with the COVID-19 virus.¹⁵ The Chinese Center for Disease Control and Prevention reported a case fatality rate (CFR) of 2.3% (1023 deaths among 44,672 confirmed cases). The case fatality rate was as high as 7.3% in diabetes and 6.0% in hypertension.¹⁶ A study done by Zhang et. Al. showed a positive correlation between knowledge, attitude and practices of healthcare workers during COVID-19 pandemic.¹⁷

This paper aims to identify factors affecting mental health outcomes of healthcare workers during Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center.

Research Question

What are the factors affecting mental health outcomes among healthcare workers during Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center?

Population (P) – Healthcare workers employed at Cardinal Santos Medical Center with direct patient exposure during Coronavirus Disease (COVID-19) pandemic

Exposure (E) – Demographic data, clinical factors, psychological services availed, Knowledge on Coronavirus Disease (COVID-19)

Outcome (O) – Mental health outcomes

Methods (M) – Cross-sectional study

Objective of the study

General Objectives

This paper aims to identify the factors affecting mental health outcome of healthcare workers during Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center from March 2020 to September 2020.

Specific Objectives

This paper specifically aims to:

- 1) Determine the prevalence of mental health outcome in terms of depression (measured by the 9-item Patient Health Questionnaire (PHQ-9)), anxiety (measured by 7- item

Generalized Anxiety Disorder (GAD-7)), and distress (measured by 22-item Impact of Event Scale-Revised (IES- R)) of healthcare workers during Coronavirus Disease (COVID-19) pandemic.

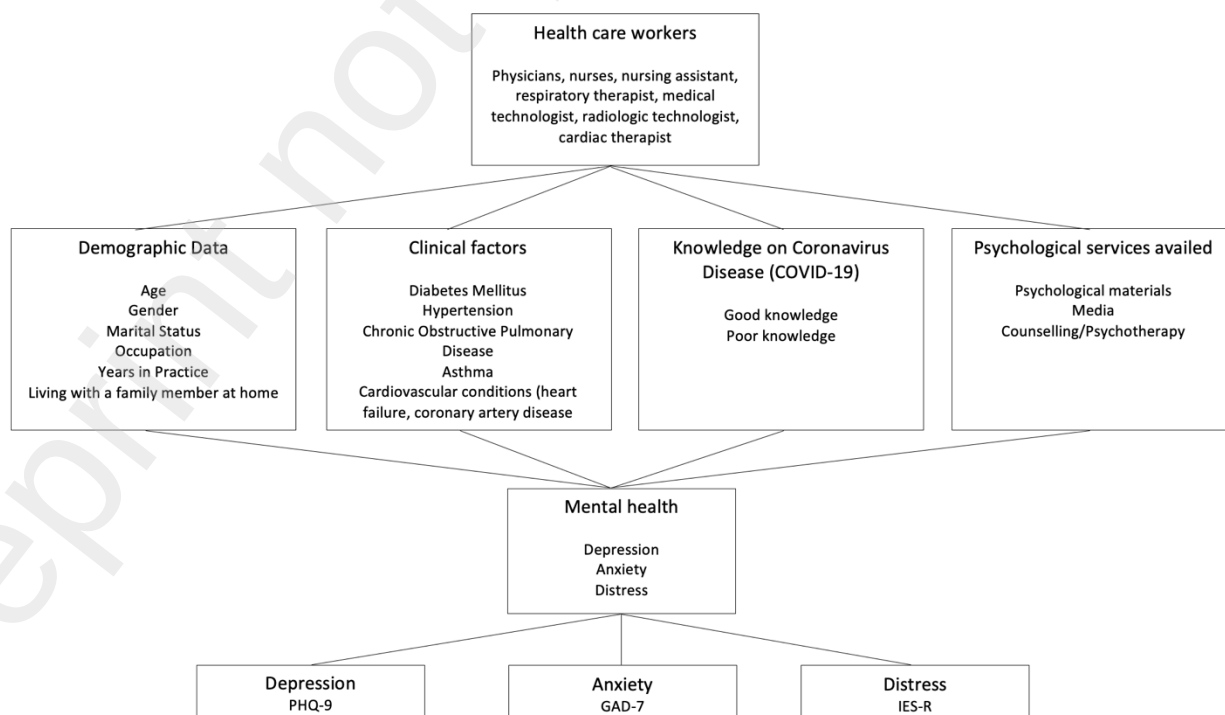
- 2) Identify the factors (demographic data, clinical factors, knowledge on Coronavirus Disease COVID-19 and psychological services availed) affecting mental health and its association to mental health outcome.

Significance of the study

Depression anxiety and distress may be viewed as normal emotional reactions to stress in the face of a pandemic.¹⁸ Healthcare workers are at risk of emotional strain and physical exhaustion from the provision of care to growing numbers of rapidly deteriorating patients; exposure to critical illness or death of their co-workers; and moral dilemmas in decision making around provision of care with limited resources¹⁹⁻²⁰

There is a need to recognize mental health as an important component of our health care workers' well-being, and provide them with mental support whether through face-to-face counseling or comparable support through digital platforms such as cell phone interfaces to extend their immediate efficiency and to better protect their overall sense of wellbeing in the long term.¹³ Building a culture of organizational resilience may help to reduce the psychological manifestations requiring immediate attention and treatment.

Conceptual Framework



Scope and Limitations

This is a cross-sectional study and the researchers involving only healthcare workers with direct contact to a patient with confirmed or suspected case Coronavirus Disease (COVID-19) at Cardinal Santos Medical Center from March 2020 to September 2020 employed during the study period. The following healthcare workers were included: physicians, nurses, nursing assistant, respiratory therapist, medical technologist, radiologic technologist, cardiac therapist with direct contact to a patient with confirmed or suspected case of Coronavirus Disease (COVID-19). The mental health outcomes include depression, anxiety and distress were measured using the 9-item Patient Health Questionnaire (PHQ-9), the 7-item Generalized Anxiety Disorder (GAD-7), and the 22-item Impact of Event Scale-Revised (IES- R) respectively. Factors affecting of metal health were identified in terms of demographic data, clinical factors, knowledge of Coronavirus Disease (COVID-19) and psychological services availed and its association to mental health outcome were determined.

Definition of Terms

1. **Mental health outcome** – important part of overall health and well-being which includes emotional, psychological, and social wellbeing.²² In this study, depression, anxiety, and distress were measured using The 9-item Patient Health Questionnaire (PHQ-9), the 7-item Generalized Anxiety Disorder (GAD-7), and the 22-item Impact of Event Scale-Revised (IES- R) respectively.
2. **Depression** – characterized by feelings of sadness and/or a loss of interest in activities once has previously enjoyed.²³ In this study, PHQ-9 was used to measure the severity of depression, with the total scores categorized as follows: minimal/no depression (1-4), Mild depression (5–9), moderate depression (10–14), or severe depression (15–21) (Kocalevent et al., 2013).²⁴
3. **Anxiety** - characterized by feelings of tension, worried thoughts and physical changes.²⁵ In this study, GAD-7 was used to evaluate the severity of anxiety The total scores are categorized as follows: minimal/no anxiety (0–4), mild anxiety (5–9), moderate anxiety (10–14), or severe anxiety (15–21) (Löwe et al., 2008).²⁶
4. **Distress** – characterized by excessive fatigue or exhaustion.²⁷ In this study, the IES-R is was used to measure the response to a specific stressful life event. The event used for this questionnaire was the occurrence of COVID-19. The total scores are categorized as follows: subclinical (0–8), mild distress (9–25), moderate distress (26–43), and severe distress (44–88) (Daniel and Weiss, 2007).²⁸

5. **Healthcare worker** – a worker involved in Coronavirus disease (COVID-19) prevention and treatment and exposed to with confirmed or suspected cases through patient screening, inspection, testing, transport, treatment, nursing, specimen collection, pathogen detection, pathologic examination, or pathologic anatomy of medical and healthcare professional and technical personnel.¹⁵ In this study, the following were included: physicians, nurses, nursing assistant, respiratory therapist, medical technologist, radiologic technologist, cardiac therapist with direct contact to a patient with confirmed or suspected case of Coronavirus Disease (COVID-19).
6. **Coronavirus Disease (COVID-19) pandemic** – a disease caused by the novel corona virus 2019 that was first reported and then became widespread within Wuhan, the capital city of Hubei Province of China. This disease rapidly spread throughout China and elsewhere, becoming a global health emergency.²⁹
7. **Demographic data** – In this study, these includes age (years), gender (male or female), marital status (single, married), occupation (physicians, nurses, nursing assistant, respiratory therapist, medical technologist, radiologic technologist, cardiac therapist), years in practice, living with a family member or non-family member with or without comorbidities.
8. **Clinical factors** – Preexisting conditions such as diabetes, heart disease, preexisting heart condition, chronic obstructive lung disease, asthma, identified by the World Health Organization that predisposes a to become severely ill with the Coronavirus Disease (COVID-19).²⁹
9. **Psychological care** – includes psychological materials, psychological resources available through, and counseling or psychotherapy to support the holistic wellbeing of an individual.¹³
10. **Knowledge on Coronavirus disease (COVID-19)** – This is a 10-item true or false question regarding knowledge of healthcare workers on Coronavirus disease (COVID-19) based on Question and Answer about COVID-19 in the webpage of WHO³⁰ and was lifted from the study of Giao, H. et. Al and the Center for Disease Control on COVID 19.³³⁻³⁴ Each correct answer in relation to the knowledge of COVID- 19 was given one point. The total knowledge score for the HCWs varied between 0 (with no correct answer) and 10 (for all correct answers), and a cut off level of <7 was evaluated as poor knowledge, and >= 7 indicated good knowledge.³²

2. Methodology

Research Design

This is a cross-sectional study of Factors affecting mental health outcomes of healthcare workers during Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center.

Sampling Design

Inclusion Criteria:

Healthcare workers with direct contact to patients with Coronavirus Disease (COVID-19) at Cardinal Santos Medical Center from March 2020 to September 2020 employed during the study period.

Exclusion Criteria:

Volunteer health workers, healthcare workers with less than 3 month work experience, healthcare workers who did not have direct contact to a patient with suspected or known Coronavirus Disease (COVID-19), healthcare worker who did not give consent, did not complete questionnaire or gave no response after 3 consecutive follow-up.

Sample Size

Using OpenEpi version 3, sample size was computed using 95% confidence interval, a population size of 157 based on the 263 healthcare workers (40 physicians, 150 nurses and nursing assistant, 26 respiratory therapist, 14 medical technologist, 10 radiologic technologist, 23 cardiac therapist) with direct contact to Coronavirus Disease (COVID-19) patients at Cardinal Santos Medical Center from March 2020 to September 2020. All that passed the inclusion criteria and exclusion criteria will be requested to participate. A hypothesized frequency of 50% was used based on the prevalence of mental health outcomes in terms of depression, anxiety and distress from the study of Lai, et. Al, and Kang et. Al. .¹²⁻¹³

Sample Size for Frequency in a Population

Population size(for finite population correction factor or fpc)(N): 263
Hypothesized % frequency of outcome factor in the population (p): 50%+/-5
Confidence limits as % of 100(absolute +/- %)(d): 5%
Design effect (for cluster surveys-DEFF): 1

Sample Size(n) for Various Confidence Levels

ConfidenceLevel(%)	Sample Size
95%	157
80%	102
90%	134
97%	169
99%	189
99.9%	212
99.99%	225

Equation

Sample size $n = [DEFF * Np(1-p)] / [(d^2 / Z^2_{1-\alpha/2} * (N-1) + p * (1-p)]$

Results from OpenEpi, Version 3, open source calculator--SSPropor
Print from the browser with ctrl-P
or select text to copy and paste to other programs.

Figure 1. Sample size computation

Method

The research protocol was technically approved by Cardinal Santos Medical Center's Institutional Review Board. Initiation of study was done after permission was granted from the Chairman of Human Resource Management, Head of Ancillary Division, and Nursing Research Team of Cardinal Santos Medical Center.

A written questionnaire was distributed thru the section heads of the Residents Training officer, Nursing Research Team, Head of Ancillary (which includes the Pulmonary section, Cardiac section, Laboratory section and Radiologic Section) and results were collected within 1 week. A digital questionnaire via google form was available. Department and Section Heads sent this thru email blast to all healthcare workers involved. The researcher followed up the department/section heads three times before excluding the participant who did not answer the questionnaire before dropping off the respondent. See attached Appendix C for sample of the questionnaire.

Data were encoded by the researcher and statistician. The researcher, adviser and statistician only have access to both hard copy and electronic copy of the results of the research. Hard copy was stored on a cabinet with lock and key. The electronic copy was stored in a password protected document. After 1 year, hard copy will be shredded and electronic data will be deleted.

After data collection, those who were noted to have severe depression, anxiety and/or distress were recommended to have a psychiatric referral for evaluation.

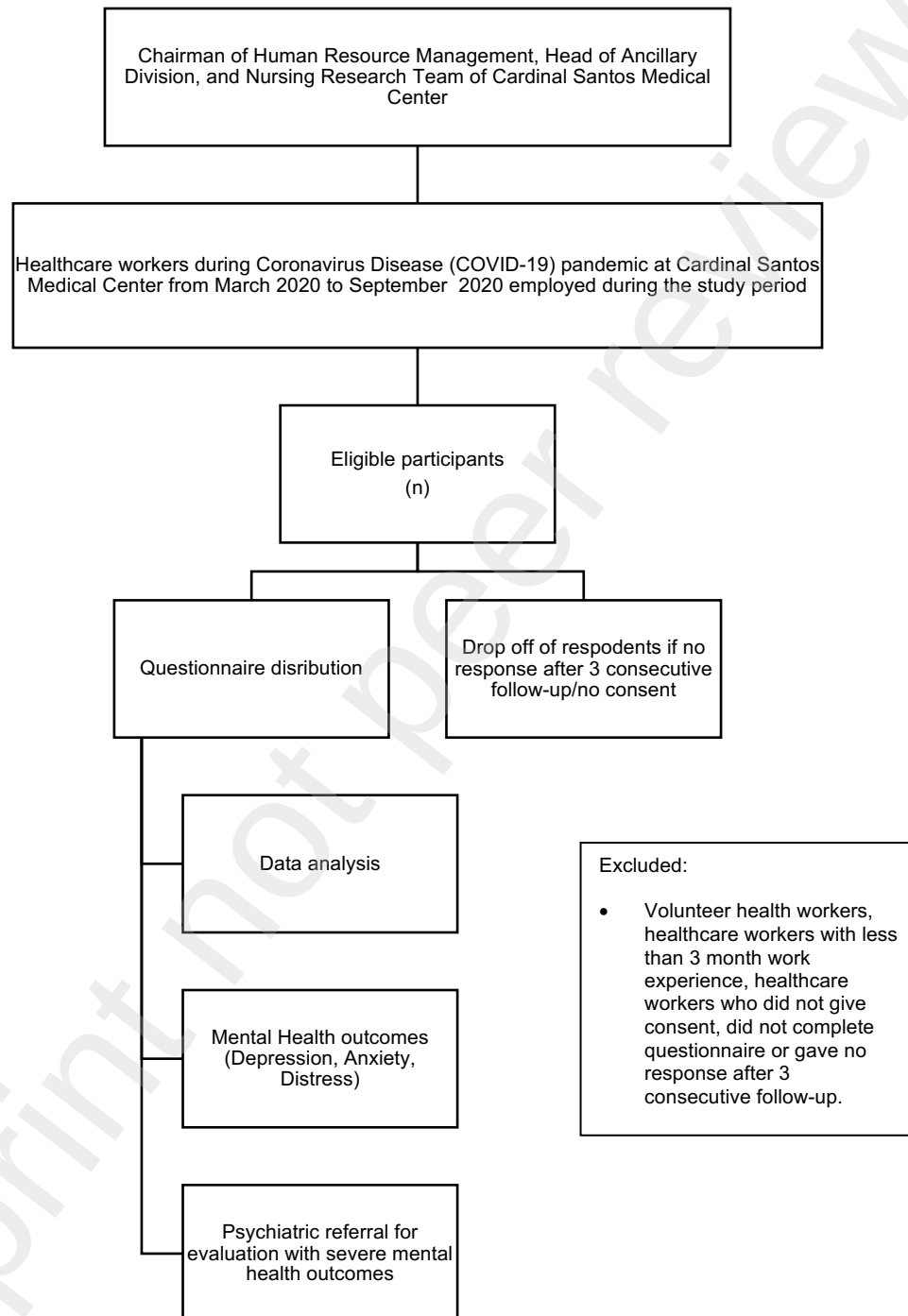


Figure 2. Flow chart/algorithm of procedure

Statistical Analysis

Study variables were summarized as percentages for categorical variables and median (IQR) for continuous variables.

The significant differences in severity of symptoms among different subgroups were determined using Fisher's exact test for categorical variables and Mann-Whitney test (2 groups) or Kruskal-Wallis test (more than 2 groups), for continuous variables.

Univariable and multivariable binomial logistic regression model was used to determine the crude and adjusted association of demographic and clinical factors, psychological services availed and knowledge on COVID19 with the prevalence of depression, anxiety, and distress symptoms. Odds ratios were computed to determine strength of associations.

Ethical Consideration

The clinical protocol was reviewed and approved by the Research Ethics Review Committee (RERC) of Cardinal Santos Medical Center. The researchers ensured that all information gathered were kept strictly confidential and used solely for the purpose of the study. The study involved participants to answer a questionnaire on the Factors affecting mental health outcomes of healthcare workers during Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center. Identification numbers were used to assure and protect the privacy of patients. An informed consent was given is attached together with the questionnaire. No conflict of interest.

3. Results

Demographic profile

In this study, purposeful sampling was done and questionnaires were distributed to different section heads and were given to healthcare workers at Cardinal Santos Medical Center with direct contact to patients with Coronavirus Disease (COVID-19) from March 2020 to September 2020. A total of 172 healthcare workers (from an initial sample size of 157) participated in the study. 64.5% were females, most within the age range 20-30 years old (63.4%) and were single (81.4%). 44.8% of the health workers were nurses, 18.6% were physicians, and 14.5% were nursing assistants. 68.6% were working in practice for less than 5 years. On average, each healthcare worker lives with 3 relative/non-relative. About 69.2% were living with their relative, while 21.5% were living together with their non-relative. More than one-third were living with relatives with comorbidities and only 2.9% were living with non-relative with comorbidities. Although 76.7% of healthcare workers had no comorbidities, the most common were asthma (8.1%), hypertension (7%) and other pre-existing conditions (5.2%). The respondents were also asked about the psychological support that they had

received or availed and majority (79.7%) did not receive any psychological support. For those who received any support, the most common were Psychological resources available through media (13.4%), Psychological materials (4.7%), and counselling or therapy (4.7%). Knowledge about COVID19 was also measured among healthcare workers, and 70.9% had good knowledge on the disease.

Table 1. Demographic profile, clinical factors, psychological services availed and knowledge on COVID-19 of healthcare workers during Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center (n=172)

Characteristic	n (%)
Age group	
20-30	109 (63.4%)
31-40	39 (22.7%)
>= 41	24 (13.9%)
Sex	
Male	61 (35.5%)
Female	111 (64.5%)
Marital Status	
Single	140 (81.4%)
Married	32 (18.6%)
Occupation	
Physician	32 (18.6%)
Nurse	77 (44.8%)
Nursing assistant	25 (14.5%)
Respiratory therapist	16 (9.3%)
Radiologic technologist	13 (7.6%)
Medical technologist	5 (2.9%)
Cardiac therapist	4 (2.3%)
Years in practice	
<5years	118 (68.6%)
5-20 years	33 (19.2%)
>20years	21 (12.2%)
Number of people living with you at home	3 (1-5)
Living with relative	119 (69.2%)
Living with non-relative	37 (21.5%)
Living with relative with comorbidities	63 (36.6%)
Living with non-relative with comorbidities	5 (2.9%)
Comorbidities	
None	132 (76.7%)
Asthma	14 (8.1%)

Hypertension	12 (7%)
Other preexisting conditions	9 (5.2%)
Diabetes mellitus	4 (2.3%)
Other cardiovascular conditions	2 (1.2%)
Chronic obstructive pulmonary disease	1 (0.6%)
Psychological support	
None	137 (79.7%)
Psychological resources available through media	23 (13.4%)
Psychological materials	8 (4.7%)
Counselling or therapy	8 (4.7%)
Others	4 (2.3%)
Knowledge	
Good	122 (70.9%)
Poor	50 (29.1%)

Prevalence of severity of depression, anxiety and distress symptoms

Tables 2.1- Table 2.6 shows the prevalence and severity of depression, anxiety, and distress symptoms among the healthcare workers according to their age, sex, marital status, occupation, living with relative or non-relative at home, and knowledge on COVID19. Prevalence of depression, anxiety and distress symptoms among all healthcare workers was 57%, 47.7%, 65.1%, respectively. Of the total (Table 2.1), most had mild symptoms of depression (32%) and only 2.3% had severe symptoms of depression. 27.9% had mild symptoms of anxiety and 7% had severe symptoms of anxiety. 40.1% had mild symptoms of distress and 5.8% had severe symptoms distress.

Age

Across age groups (Table 2.1), healthcare workers aged over 41 had no symptoms of depression (62.5%), and none had moderately severe or severe symptoms among those aged 31 and above. On the other hand, 13.8% and 3.7% of the those in age 20-30 had moderately severe and severe depression symptoms, respectively. Results also show that severity of depression symptoms significantly varies across age groups (p -value = 0.004), and depression scores significantly decreases as age groups become older (p =.0001).

Regarding anxiety symptoms, percentage of no symptoms increases by age group. More young healthcare workers had moderate and severe symptoms compared to older age groups. Results also show that severity of anxiety symptoms significantly varies across age groups (p -value = 0.02). Anxiety scores significantly differ among age groups, where scores decrease as age group increases (p =0.0035).

At least half of the healthcare workers aged 31 and above had mild distress symptoms, while higher proportion of young workers had moderate (23.9%) and severe distress symptoms (8.3%).

Table 2.1. Severity of depression, anxiety and distress symptoms by age

	Total		Age group						
Severity category	n (%)	CI	20-30	CI	31-40	CI	41-50	CI	p-value
PHQ-9, depression symptoms									
Severity									0.004
None	74 (43%)	35.8-50.6	36 (33%)	24.8-42.5	23 (59%)	42.8-73.4	15 (62.5%)	41.6-79.6	
Mild	55 (32%)	25.4-39.4	34 (31.2%)	23.1-40.6	14 (35.9%)	22.3-52.2	7 (29.2%)	14.3-50.5	
Moderate	24 (14%)	9.5-20	20 (18.4%)	12.1-26.9	2 (5.1%)	1.2-18.8	2 (8.3%)	2-28.7	
Moderately severe	15 (8.7%)	5.3-14	15 (13.8%)	8.4-21.7	0		0		
Severe	4 (2.3%)	0.9-6.1	4 (3.7%)	1.4-9.5	0		0		
Score	6 (2-9.5)		7 (4-12)		4 (0-6)		1.5 (0-7)		0.0001
GAD-7, anxiety									
Severity									0.02
None	90 (52.3%)	44.8-59.8	47 (43.1%)	34.1-52.7	28 (71.8%)	55.5-83.8	15 (62.5%)	41.6-79.6	
Mild	48 (27.9%)	21.7-35.1	32 (29.4%)	21.5-38.7	9 (23.1%)	12.3-39.1	7 (29.2%)	14.3-50.5	
Moderate	22 (12.8%)	8.5-18.7	19 (17.4%)	11.3-25.8	2 (5.1%)	1.2-18.8	1 (4.2%)	0.6-25.4	
Severe	12 (7%)	4-11.9	11 (10.1%)	5.6-17.4	0		1 (4.2%)	0.6-25.4	
Score	4 (1-8)		6 (2-10)		3 (0-5)		3 (2-0)		0.0035
IES-R, distress symptoms									
Severity									0.076
Normal	60 (34.9%)	28.1-42.4	37 (33.9%)	25.6-43.4	16 (41%)	26.6-57.2	7 (29.2%)	14.3-50.5	
Mild	69 (40.1%)	33-47.7	37 (33.9%)	25.6-43.4	20 (51.3%)	35.7-66.6	12 (50%)	30.5-69.5	
Moderate	33 (19.2%)	13.9-25.8	26 (23.9%)	16.7-32.8	3 (7.7%)	2.4-21.7	4 (16.7%)	6.2-37.6	
Severe	10 (5.8%)	3.1-10.5	9 (8.3%)	4.3-15.2	0		1 (4.2%)	0.6-25.4	
Score	13 (6-24)		13 (6-31)		12 (5-18)		14 (7-21)		0.2198

Sex

Prevalence of depression symptoms was higher among females (60.4%) than males (50.8%) (Table 2.2). About the same proportion of males and females had mild and moderately severe depression symptoms, while more females had moderate and severe depression symptoms.

Although almost the same proportion of males (52.5%) and females (52.3%) had no anxiety symptoms, more males (32.8%) had mild anxiety symptoms while more females had moderate (15.3%) and severe anxiety symptoms (7.2%).

In terms of distress symptoms, more females experienced mild (45.1%) and severe (8.1%) symptoms while more males experienced moderate (23%) symptoms. Results

show that severity of depression, anxiety and distress was not statistically significant among male and female.

Marital Status

Prevalence of depression symptoms was higher among singles than married healthcare workers (Table 2.2). Although the same proportion between singles and married had mild depression symptoms, no married healthcare worker had moderately severe or severe depression symptoms. On the other hand, 10.7% and 2.9% of singles experienced moderately severe and severe depression symptoms.

Married healthcare workers had more mild anxiety symptoms, but more singles had moderate and severe anxiety symptoms.

In terms of distress symptoms, more than half of the married group had mild distress symptoms and 15.6% had moderate distress symptoms and no one had severe symptoms. On the other hand, at least one-third of the singles had mild, 20% had moderate and 7% had severe distress symptoms.

Results show that severity of depression, anxiety and distress was not statistically significant in terms of marital status.

Table 2.2. Severity of depression, anxiety and distress symptoms by sex and marital status

	Sex					Marital Status				
Severity category	Male	CI	Female	CI	p-value	Single	CI	Married	CI	p-value
PHQ-9, depression symptoms										
Severity					0.442					0.206
None	30 (49.2%)	36.8-61.7	44 (39.6%)	30.9-49.1		56 (40%)	32.1-48.4	18 (56.3%)	38.6-72.4	
Mild	20 (32.8%)	22.1-45.6	35 (31.5%)	23.5-40.8		45 (32.1%)	24.9-40.4	10 (31.3%)	17.5-49.4	
Moderate	6 (9.8%)	4.4-20.4	18 (16.2%)	10.4-24.4		20 (14.3%)	9.4-21.2	4 (12.5%)	4.7-29.4	
Moderately severe	5 (8.2%)	3.4-18.4	10 (9%)	4.9-16		15 (10.7%)	6.5-17.1	0		
Severe	0		4 (3.6%)			4 (2.9%)	1.1-7.4	0		
Score	5 (2-8)		6 (2-11)		0.1438	6 (3-10)		3 (0-6.5)		0.0032
GAD-7, anxiety										
Severity					0.499					0.779
None	32 (52.5%)	39.8-64.8	58 (52.3%)	42.9-61.5		72 (51.4%)	43.1-59.7	18 (56.3%)	38.6-72.4	
Mild	20 (32.8%)	22.1-45.6	28 (25.2%)	17.9-34.2		38 (27.1%)	20.4-35.2	10 (31.3%)	17.5-49.4	
Moderate	5 (8.2%)	3.4-18.4	17 (15.3%)	9.7-23.4		19 (13.6%)	8.8-20.4	3 (9.4%)	3-25.9	
Severe	4 (6.6%)	2.4-16.4	8 (7.2%)	3.6-13.9		11 (7.9%)	4.4-13.7	1 (3.1%)	0.4-19.8	
Score	4 (0-7)		4 (1-9)		0.4009	4 (1-9)		3.5 (0-7)		0.2889
IES-R, distress symptoms										
Severity					0.051					0.257
Normal	27 (44.3%)	32.2-57	33 (29.7%)	21.9-39		50 (35.7%)	28.1-44.1	10 (31.3%)	17.5-49.4	
Mild	19 (31.2%)	20.7-44	50 (45.1%)	36-54.5		52 (37.1%)	29.5-45.5	17 (53.1%)	35.8-69.8	
Moderate	14 (23%)	14-35.3	19 (17.1%)	11.1-25.4		28 (20%)	14.1-27.5	5 (15.6%)	6.5-33	
Severe	1 (1.6%)	0.2-11	9 (8.1%)	4.2-15		10 (7.1%)	3.9-12.8	0		
Score	11 (3-20)		14 (7-26)		0.1013	13 (5.5-26.5)		13.5 (7.5-21)		0.8314

Occupation

Among different types of occupation (Table 2.3), prevalence of depression symptoms was highest among therapists (cardiac and pulmonary) and technologists (medical and radiologic) (65.8%). Higher proportion of nursing assistant, therapists (cardiac and pulmonary), and technologists (medical and radiologic) experienced moderately severe and severe depression symptoms than physicians and nurses.

Prevalence of anxiety symptoms was again highest among therapists (cardiac and pulmonary), and technologist (medical and radiologic) (52.6%), and had moderate (21.1%) and severe (13.2%) symptoms.

The same pattern is also observed in terms of distress symptoms, where therapists (cardiac and pulmonary) and technologists (medical and radiologic) (73.7%) had the highest prevalence, and mostly had experienced mild (39.5) and moderate (26.3%) symptoms.

Results show that severity of depression, anxiety and distress was not statistically significant between occupation (physician, nurses, nursing assistant, cardiac/pulmonary therapists and medical/radiologic technologist).

Table 2.3. Severity of depression, anxiety and distress symptoms by occupation

	Occupation								
Severity category	Physicians	CI	Nurses	CI	Nursing Assistants	CI	Therapists and Technologists	CI	p-value
PHQ-9, depression symptoms									
Severity									0.281
None	16 (50%)	33-67	31 (40.3%)	29.8-51.7	14 (56%)	36.1-74.1	13 (34.2%)	20.8-50.8	
Mild	9 (28.1%)	15.1-46.3	29 (37.7%)	27.5-49.1	5 (20%)	8.4-40.6	12 (31.6%)	18.7-48.1	
Moderate	6 (18.8%)	8.5-36.4	10 (13%)	7.1-22.6	2 (8%)	1.9-27.7	6 (15.8%)	7.1-31.3	
Moderately severe	0		5 (6.5%)	2.7-14.8	4 (16%)	6-36.4	6 (15.8%)	7.1-31.3	
Severe	1 (3.1%)	0.4-19.8	2 (2.6%)	0.6-10	0		1 (2.6%)	0.4-17	
Score	4.5 (1.5-9)		6 (3-9)		3 (1-9)		8 (3-12)		0.2051
GAD-7, anxiety									
Severity									0.487
None	19 (59.4%)	41.5-75.1	38 (49.4%)	38.3-60.5	15 (60%)	39.7-77.3	18 (47.4%)	32-63.3	
Mild	10 (31.3%)	17.5-49.4	25 (32.5%)	22.9-43.8	6 (24%)	10.9-44.8	7 (18.4%)	8.9-34.3	
Moderate	2 (6.3%)	1.5-22.4	9 (11.7%)	6.1-21.1	3 (12%)	3.8-32	8 (21.1%)	10.7-37.2	
Severe	1 (3.1%)	0.4-19.8	5 (6.5%)	2.7-14.8	1 (4%)	0.5-24.6	5 (13.2%)	5.5-28.4	
Score	3 (0-6.5)		5 (1-8)		3 (0-8)		5 (2-13)		0.1921
IES-R, distress symptoms									
Severity									0.622
Normal	14 (43.8%)	27.6-61.4	29 (37.7%)	27.5-49.1	7 (28%)	13.7-48.8	10 (26.3%)	14.6-42.7	
Mild	11 (34.4%)	19.9-52.5	31 (40.3%)	29.8-51.7	12 (48%)	29.2-67.4	15 (39.5%)	25.1-55.9	
Moderate	7 (21.9%)	10.6-39.8	12 (15.6%)	9-25.6	4 (16%)	6-36.4	10 (26.3%)	14.6-42.7	
Severe	0		5 (6.5%)	2.7-14.8	2 (8%)	1.9-27.7	3 (7.9%)	2.5-22.2	
Score	9.5 (2-22.5)		13 (6-21)		16 (8-23)		15 (8-36)		0.3502

Living with a relative or non-relative at home

A higher proportion of depression, anxiety and distress symptoms was experienced by healthcare workers living with a relative as compared with non-relative at home (Table 2.4). Higher proportion of those with living with a relative mild symptoms of depression (36.1%). Results show that severity of depression symptoms varies according living with a relative at home ($p=0.004$).

Higher proportion of those with living with a relative have mild symptoms of anxiety (31.9%), but those living with non-relative have moderate symptoms of anxiety (21.6%).

Higher proportion of those with living with a relative have mild (42.9%) and severe (5.7%) symptoms of distress, but those living with non-relative have moderate symptoms of anxiety (29.7%).

Table 2.4. Severity of depression, anxiety and distress symptoms by living with a relative or non-relative at home

Severity category	Living with relative				p-value	Living with non-relative				p-value
	Yes	CI	No	CI		Yes	CI	No	CI	
PHQ-9, depression symptoms										
Severity					0.004					0.352
None	51 (42.9%)	34.2-52	23 (43.4%)	30.6-57.1		16 (43.2%)	28.2-59.7	58 (43%)	34.8-51.5	
Mild	43 (36.1%)	27.9-45.2	12 (22.6%)	13.2-36		8 (21.6%)	11-38	47 (34.8%)	27.2-43.3	
Moderate	19 (16%)	10.4-23.8	5 (9.4%)	3.9-21		7 (18.9%)	9.2-35.1	17 (12.6%)	7.9-19.4	
Moderately severe	5 (4.2%)	1.7-9.8	10 (18.9%)	10.4-31.9		5 (13.5%)	5.6-29	10 (7.4%)	4-13.3	
Severe	1 (0.8%)	0.1-5.8	3 (5.7%)	1.8-16.4		1 (2.7%)	0.4-17.4	3 (2.2%)	0.7-6.7	
Score	6(2-9)		6(2-14)		0.1749	6(2-11)		6(2-9)		0.4127
GAD-7, anxiety										
Severity					0.089					0.229
None	63 (52.9%)	43.9-61.8	27 (50.9%)	37.5-64.2		20 (54.1%)	37.8-69.5	70 (51.9%)	43.4-60.2	
Mild	38 (31.9%)	24.1-40.9	10 (18.9%)	10.4-31.9		7 (18.9%)	9.2-35.1	41 (30.4%)	23.1-38.7	
Moderate	12 (10.1%)	5.8-17	10 (18.9%)	10.4-31.9		8 (21.6%)	11-38	14 (10.4%)	6.2-16.8	
Severe	6 (5%)	2.3-10.9	6 (11.3%)	5.1-23.2		2 (5.4%)	1.3-19.7	10 (7.4%)	4-13.3	
Score	4(0-7)		4(1-11)		0.1498	4(1-10)		4(1-8)		0.7496
IES-R, distress symptoms										
Severity					0.408					0.2
Normal	42 (35.3%)	27.2-44.4	18 (34%)	22.4-47.8		14 (37.8%)	23.6-54.5	46 (34.1%)	26.5-42.6	
Mild	51 (42.9%)	34.2-52	18 (34%)	22.4-47.8		11 (29.7%)	17.1-46.5	58 (43%)	34.8-51.5	
Moderate	19 (16%)	10.4-23.8	14 (26.4%)	16.2-40		11 (29.7%)	17.1-46.5	22 (16.3%)	10.9-23.6	
Severe	7 (5.9%)	2.8-11.9	3 (5.7%)	1.8-16.4		1 (2.7%)	0.4-17.4	9 (6.7%)	3.5-12.4	
Score	13(7-22)		13(3-31)		0.8459	13(3-28)		13(6-23)		0.9896

Living with a relative with comorbidities

A higher prevalence of depression, anxiety and distress symptoms was experienced by healthcare workers if the relative living at home have comorbidities (Table 2.5). Higher proportion of those with living with a relative with comorbidities have mild (42.9%), moderate (20.6%) symptoms of depression. Results show that severity of depression symptoms varies according living with a relative at home with comorbidities ($p=0.007$).

Higher proportion of those with living with a relative with comorbidities have mild (42.9%), moderate (12.7%) symptoms of anxiety. Results show that severity of anxiety symptoms varies according living with a relative at home with comorbidities ($p=0.009$).

Higher proportion of those with living with a relative with comorbidities have mild (46%), moderate (19%) and severe (6.3%) symptoms of distress.

Table 2.5. Severity of depression, anxiety and distress symptoms by living with a relative at home with comorbidities

Severity category	Living with relative				p-value	Living with relative with comorbidities				p-value
	Yes	CI	No	CI		Yes	CI	No	CI	
PHQ-9, depression symptoms										
Severity					0.004					0.007
None	51 (42.9%)	34.2-52	23 (43.4%)	30.6-57.1		20 (31.7%)	21.3-44.4	54 (49.5%)	40.2-58.9	
Mild	43 (36.1%)	27.9-45.2	12 (22.6%)	13.2-36		27 (42.9%)	31.1-55.5	28 (25.7%)	18.3-34.8	
Moderate	19 (16%)	10.4-23.8	5 (9.4%)	3.9-21		13 (20.6%)	12.3-32.6	11 (10.1%)	5.6-17.4	
Moderately severe	5 (4.2%)	1.7-9.8	10 (18.9%)	10.4-31.9		3 (4.8%)	1.5-14	12 (11%)	6.3-18.5	
Severe	1 (0.8%)	0.1-5.8	3 (5.7%)	1.8-16.4		(0%)	0-0	4 (3.7%)	1.4-9.5	
Score	6(2-9)		6(2-14)		0.1749	6(4-10)		5(1-9)		0.1614
GAD-7, anxiety										
Severity					0.089					0.009
None	63 (52.9%)	43.9-61.8	27 (50.9%)	37.5-64.2		25 (39.7%)	21.3-44.4	65 (59.6%)	40.2-58.9	
Mild	38 (31.9%)	24.1-40.9	10 (18.9%)	10.4-31.9		27 (42.9%)	31.1-55.5	21 (19.3%)	18.3-34.8	
Moderate	12 (10.1%)	5.8-17	10 (18.9%)	10.4-31.9		8 (12.7%)	12.3-32.6	14 (12.8%)	5.6-17.4	
Severe	6 (5%)	2.3-10.9	6 (11.3%)	5.1-23.2		3 (4.8%)	1.5-14	9 (8.3%)	6.3-18.5	
Score	4(0-7)		4(1-11)		0.1498	5(2-9)		3(0-7)		0.0959
IES-R, distress symptoms										
Severity					0.408					0.548
Normal	42 (35.3%)	27.2-44.4	18 (34%)	22.4-47.8		18 (28.6%)	21.3-44.4	42 (38.5%)	40.2-58.9	
Mild	51 (42.9%)	34.2-52	18 (34%)	22.4-47.8		29 (46%)	31.1-55.5	40 (36.7%)	18.3-34.8	
Moderate	19 (16%)	10.4-23.8	14 (26.4%)	16.2-40		12 (19%)	12.3-32.6	21 (19.3%)	5.6-17.4	
Severe	7 (5.9%)	2.8-11.9	3 (5.7%)	1.8-16.4		4 (6.3%)	1.5-14	6 (5.5%)	6.3-18.5	
Score	13(7-22)		13(3-31)		0.8459	16(7-26)		11(3-22)		0.0531

Knowledge on COVID-19

Based on knowledge on COVID-19, higher prevalence of depressions, anxiety and distress symptoms was experienced by healthcare workers with poor knowledge compared with those with good knowledge (Table 2.6). Higher proportion of those with poor knowledge had mild (46%) and moderately severe (14%) depression symptoms. On the other hand, higher proportion of those with good knowledge had moderate symptoms of depression (18%). Results show that severity of depression symptoms varies according to their knowledge on COVID-19 ($p=0.007$).

Healthcare workers with poor knowledge on COVID19 had higher proportion of mild, moderate and severe anxiety symptoms. In addition, higher proportion of those with poor knowledge had mild and severe distress symptoms, while higher proportion of those with good knowledge had moderate distress symptoms.

Table 2.6. Severity of depression, anxiety and distress symptoms by knowledge on coronavirus disease (COVID-19)

	Knowledge				
Severity category	Poor	CI	Good	CI	p-value
PHQ-9, depression symptoms					
Severity					0.007
None	17 (34%)	22.1-48.3	57 (46.7%)	38-55.7	
Mild	23 (46%)	32.6-60	32 (26.2%)	19.1-34.8	
Moderate	2 (4%)	1-14.9	22 (18%)	12.1-26	
Moderately severe	7 (14%)	6.7-26.8	8 (6.6%)	3.3-12.7	
Severe	1 (2%)	0.3-13.3	3 (2.5%)	0.8-7.4	
Score	6.5 (3-9)		6 (2-10)		0.284
GAD-7, anxiety					
Severity					0.678
None	23 (46%)	32.6-60	67 (54.9%)	45.9-63.6	
Mild	15 (30%)	18.8-44.2	33 (27.1%)	19.8-35.7	
Moderate	8 (16%)	8.1-29.1	14 (11.5%)	6.9-18.5	
Severe	4 (8%)	3-19.7	8 (6.6%)	3.3-12.7	
Score	5 (2-9)		4 (1-7)		0.2868
IES-R, distress symptoms					
Severity					0.37
Normal	14 (28%)	17.2-42.2	46 (37.7%)	29.5-46.7	
Mild	24 (48%)	34.4-61.9	45 (36.9%)	28.7-45.9	
Moderate	8 (16%)	8.1-29.1	25 (20.5%)	14.2-28.7	
Severe	4 (8%)	3-19.7	6 (4.9%)	2.2-10.6	
Score	16 (7-23)		12.5 (5-25)		0.3365

Univariate and Multivariate analysis of factors associated with depression, anxiety and distress

Table 3 -Table 5 shows the association of demographic and clinical factors, psychological services availed and knowledge on COVID19 with prevalence of depression, anxiety and distress symptoms using univariable and multivariable binomial logistic regression model.

Depression symptoms

Univariate analysis in Table 3 shows that age categories, years of experience, living with relatives with comorbidities, and those with asthma are significantly associated with prevalence of depression symptoms.

Twenty variables including demographic and clinical factors, psychological support and knowledge on COVID-19 were assessed using stepwise logistic regression (backward selection) to determine the significant predictors of depression symptoms.

After multivariate analysis, only age group and living with relative with comorbidities have statistically significant association with prevalence of depression symptoms. Taking these into account, the adjusted odds ratio shows that healthcare workers aged 31-40 years and 41 years and above are 70% less likely to have depression symptoms (OR 0.278, CI 95%, 0.11-0.72 $p=0.008$ and OR 0.273, CI 95%, 0.12-0.61, $p=0.002$) compared to younger health workers between 20 and 30 years old. Those living with relatives with comorbidities are 2.7 times more likely to experience depression symptoms (OR 2.731, CI 95%, 1.36-5.5, $p=0.005$).

The results also showed that therapists (cardiac and medical)/technologists (pulmonary radiologic) and nurses are 1.92 times and 1.48 times more likely to develop symptoms of depression as compared to other occupation respectively; females are 1.47 times more likely to develop symptoms of depression as compared to males; married healthcare workers are 1.92 times more likely to develop symptoms of depression as compared to single; asthmatics are 5.02 times more likely to develop symptoms of depression; those who received psychological support are more likely to develop symptoms of depression. However, these are not statistically significant.

Healthcare workers with more than 5-20 years of practice and >20 years of practice are 26% and 69.2% less likely to develop symptoms of depression as compared to those with <5 years of practice. Healthcare workers with good knowledge of COVID-19 are 41.3% are less likely to have symptoms of depression. However, these are not statistically significant.

Table 3. Results of the binomial logistic regression analysis for depression symptoms using PHQ-9

PHQ-9, depression symptoms	Univariable Analysis		Multivariable Analysis	
	Crude Odds Ratio	p-value	Adjusted Odds Ratio	p-value
Occupation				
Physicians	1			
Nurses	1.484 (0.65-3.4)	0.351		
Nursing assistant	0.786 (0.27-2.25)	0.653		
Therapists/technologists	1.923 (0.73-5.04)	0.184		
Age group				
20-30	1			
31-40	0.343 (0.16-0.73)	0.005*	0.278 (0.11-0.72)	0.008*
41-50	0.296 (0.12-0.74)	0.009*	0.273 (0.12-0.61)	0.002*
Sex				
Male	1			

Female	1.474 (0.79-2.77)	0.228		
Marital Status				
Single	1			
Married	1.929 (0.89-4.19)	0.097		
Occupation				
Years in practice				
<5years	1			
5-20 years	0.74 (0.34-1.61)	0.448		
>20years	0.308 (0.12-0.82)	0.019*		
Number of people living with you	1.056 (0.94-1.19)	0.379		
Living with relative	1.022 (0.53-1.96)	0.947		
Living with non-relative	0.989 (0.47-2.06)	0.976		
Living with relative with comorbidities	2.131 (1.12-4.04)	0.02*	2.731 (1.36-5.5)	0.005
Living with non-relative with comorbidities	0.493 (0.08-3.03)	0.445		
Comorbidities				
Diabetes mellitus	0.75 (0.1-5.45)	0.776		
Hypertension	0.515 (0.16-1.69)	0.274		
Asthma	5.023 (1.09-23.18)	0.039*		
Other cardiovascular conditions	0.753 (0.05-12.23)	0.842		
Other preexisting conditions	1.543 (0.37-6.39)	0.549		
None	0.646 (0.31-1.35)	0.244		
Psychological support				
Psychological materials	1.272 (0.29-5.5)	0.747		
Psychological resources available through media	0.979 (0.4-2.37)	0.962		
Counselling or therapy	5.615 (0.68-46.68)	0.11		
None	0.736 (0.34-1.58)	0.432		
Knowledge				
Poor	1			
Good	0.587 (0.3-1.17)	0.128		

Anxiety symptoms

Univariate analysis in Table 4 shows that age categories and living with relatives with comorbidities are significantly associated with prevalence of anxiety symptoms.

Twenty variables including demographic and clinical factors, psychological support and knowledge on COVID-19 were assessed using stepwise logistic regression (backward selection) to determine the significant predictors of anxiety symptoms.

After multivariable analysis, age group and living with a relative with comorbidities were found to be statistically associated with prevalence of anxiety symptoms. Considering both of these variables, the adjusted odds ratios show that healthcare workers age 31-40 years has 80% less likely to experience anxiety symptoms (OR 0.212, CI 95%, 0.09-0.5, $p<0.001$), and those age 41 and above have 73% less likely to have anxiety symptoms (OR 0.37, CI 95%, 0.14-0.98, $p<0.045$). Those living with relatives with comorbidities are 2.9 times more likely to have anxiety symptoms (OR 2.889, CI 95%, 1.46-5.73, $p=0.002$).

The results also showed that therapists (cardiac and medical)/technologists (pulmonary radiologic) and nurses are 1.92 times and 1.48 times more likely to develop symptoms of anxiety respectively; females are 1.47 times more likely to develop symptoms of anxiety as compared to males; married healthcare workers are 1.92 times more likely to develop symptoms of anxiety as compared to single; asthmatic healthcare workers are 5.02 times more likely to develop symptoms of anxiety; those who received psychological support are more likely to develop symptoms of anxiety. However, these are not statistically significant.

Healthcare workers with good knowledge are 30.1% less likely to have symptoms of anxiety but are not statistically significant.

Table 4. Results of the binomial logistic regression analysis for anxiety symptoms using GAD-7

GAD-7, anxiety symptoms	Univariable Analysis		Multivariable Analysis	
	Crude Odds Ratio	p-value	Adjusted Odds ratio (CI)	p-value
Occupation				
Physicians	1			
Nurses	1.5 (0.65-3.46)	0.341		
Nursing assistant	0.974 (0.34-2.83)	0.962		
Therapists/technologists	1.624 (0.63-4.2)	0.317		
Age group				
20-30				
31-40	0.298 (0.13-0.66)	0.003*	0.212 (0.09-0.5)	<0.001*
41-50	0.455 (0.18-1.13)	0.089*	0.37 (0.14-0.98)	0.045*
Sex				
Male	1			
Female	1.008 (0.54-1.88)	0.979		
Marital Status				
Single	1			
Married	1.214 (0.56-2.63)	0.623		
Years in practice				

<5years				
5-20 years	1.329 (0.61-2.88)	0.472		
>20years	0.681 (0.26-1.77)	0.43		
Number of people living with you	1.028 (0.91-1.16)	0.651		
Living with relative	0.923 (0.48-1.76)	0.809		
Living with non-relative	0.915 (0.44-1.9)	0.812		
Living with relative with comorbidities	2.273 (1.22-4.23)	0.01*	2.889 (1.46-5.73)	0.002*
Living with non-relative with comorbidities	0.725 (0.12-4.45)	0.728		
Comorbidities				
Diabetes mellitus	1.1 (0.15-7.99)	0.925		
Hypertension	0.77 (0.23-2.53)	0.666		
Asthma	1.514 (0.5-4.56)	0.462		
Other cardiovascular conditions	1.099 (0.07-17.86)	0.947		
Other preexisting conditions	1.396 (0.36-5.39)	0.628		
None	0.682 (0.33-1.39)	0.291		
Psychological support				
Psychological materials	3.474 (0.68-17.72)	0.134		
Psychological resources available through media	1.007 (0.42-2.43)	0.988		
Counselling or therapy	1.883 (0.44-8.14)	0.397		
Others	3.38 (0.34-33.15)	0.296		
None	0.828 (0.39-1.74)	0.619		
Knowledge				
Poor	1			
Good	0.699 (0.36-1.35)	0.289		

* $p < 0.005$

Distress symptoms

Univariate analysis in Table 5 shows no factor which is significantly associated with prevalence of distress symptoms.

The results also showed that therapists (cardiac and medical)/technologists (pulmonary radiologic), nursing assistant and nurses are 2.18, 2 and 1.29 times more likely to develop symptoms of distress respectively; females are 1.88 times more likely to develop symptoms of distress as compared to males; age group >41 years are 1.25 times more likely to develop symptoms of distress as compared to younger age group; health care workers with >20years of practice are 1.77 times more like to develop symptoms of distress; asthmatic and hypertensive healthcare workers are 3.48 and 1.66 times more likely to develop symptoms of distress; healthcare workers with family members with comorbidities are 1.59 times more likely to develop symptoms of distress; those who received psychological support are more likely to develop symptoms of anxiety. However, these are not statistically significant.

Married healthcare workers are 18.2% less like to develop symptoms of distress as compared to singles; healthcare workers with good knowledge are 35.7% less likely to have symptoms of distress. However, these are not statistically significant.

Table 5. Results of the binomial logistic regression analysis for distress symptoms using IES-R

IES-R, distress symptoms	Univariable Analysis	
	Crude Odds Ratio	p-value
Occupation		
Physicians	1	
Nurses	1.287 (0.56-2.97)	0.554
Nursing assistant	2 (0.65-6.12)	0.224
Therapists/technologists	2.178 (0.8-5.95)	0.129
Age group		
20-30	1	
31-40	0.739 (0.35-1.57)	0.429
41-50	1.248 (0.48-3.28)	0.653
Sex		
Male	1	
Female	1.877 (0.98-3.59)	0.057
Marital Status		
Single	1	
Married	0.818 (0.36-1.86)	0.633
Years in practice		
<5years	1	
5-20 years	0.85 (0.38-1.88)	0.689
>20years	1.768 (0.61-5.17)	0.298
Number of people living with you	1.037 (0.91-1.18)	0.57
Living with relative	0.943 (0.48-1.86)	0.866
Living with non-relative	0.849 (0.4-1.8)	0.671
Living with relative with comorbidities	1.593 (0.83-3.07)	0.164
Living with non-relative with comorbidities	0.798 (0.13-4.91)	0.808
Comorbidities		
Diabetes mellitus	0.527 (0.07-3.84)	0.528
Hypertension	1.66 (0.43-6.38)	0.46
Asthma	3.48 (0.75-16.1)	0.111
Other preexisting conditions	1.933 (0.39-9.61)	0.42
None	0.547 (0.25-1.21)	0.138

Psychological support		
Psychological materials	0.519 (0.12-2.15)	0.366
Psychological resources available through media	1.262 (0.49-3.26)	0.631
Counselling or therapy	3.933 (0.47-32.75)	0.205
Others		
None	0.823 (0.37-1.82)	0.631
Knowledge		
Poor	1	
Good	0.643 (0.31-1.32)	0.227

* $p < 0.005$

4. Discussion

This is a cross sectional study that involves 172 healthcare workers with direct contact to patients with Coronavirus Disease (COVID-19) at Cardinal Santos Medical Center. Overall, 57%, 47.7%, 65.1% respectively, of healthcare workers reported symptoms of depression, anxiety and distress. Several studies showed similar results on mental health outcome of healthcare workers during severe acute respiratory syndrome (SARS) and Middle East respiratory syndrome (MERS) outbreak, as well as during the current COVID-19 pandemic.^{6,7,8,12,13} Healthcare workers are not only under stress during pandemics, but they may also suffer psychologically long after the initial outbreak is over.⁴⁰

Age and living with a family with comorbidities were significantly associated with prevalence of depression and anxiety among healthcare workers after multivariate analysis.

Results of this research showed healthcare workers >30 years old are less likely to have symptoms of depression and anxiety as compared to younger healthcare workers (aged 21-30 years old). Younger individuals experience more adverse psychological symptoms during a pandemic as described by several studies of Ahmed et al., 2020; Gao et al., 2020; Huang and Zhao, 2020.^{35,36,37} This finding may in part be due to their caregiving role in families, who provide financial and emotional support to children or the elderly. Unpredictability of nature of their job caused by the COVID-19 pandemic among this age group could be particularly stressful.³⁸

Healthcare workers living with relative with comorbidities are more likely to have symptoms of depression and anxiety. Safety from infection was the main concern as they worried most that they might infect their families with COVID-19.^{39,40,41} Older people and people with pre-existing medical conditions (such as diabetes, heart disease, preexisting heart condition, chronic obstructive lung disease, asthma) appear to be more vulnerable to becoming severely ill with the COVID-19. Cai., et.al., cited the most important factor that helped ease the stress of the medical staff was when their family was well, not infected with COVID-19, and were not believed to be at risk of infection. Hence, reassurance of personal safety as well as providing psychological support

for medical staff might be key to encourage healthcare workers to continue working during the pandemic.³⁶

Level of knowledge on COVID-19 among health care workers significantly differs across severity of depression. Health care workers with good knowledge are 41% less likely to have depression symptoms compared to those with poor knowledge. However, there is no statistically significant association between knowledge and with prevalence of depression symptoms after binomial regression logistic analysis. Knowledge of a disease may influence healthcare workers' attitudes and practices. There are limited studies that correlated mental health outcome with knowledge.¹⁴ A study by Amin, F., et.al. showed that low-to-moderate knowledge on COVID-19 of frontline doctor was associated with depression on a multivariate analysis.⁴² Awareness of the effects of disease prevention measures with reduced numbers of reported cases reduced staff stress.³⁹

Gender was not identified as a statistically significant risk factor in this study, which is similar to the study of Huang, et. Al and Amin, et. al.^{34,39}. These results were different from Lao,, et.al and Gao, et.al. which showed female gender affecting mental health outcome.^{12,13}

Occupation was not a statistically significant risk factor, though various studies identified nurses to be more at risk for developing symptoms of depression and anxiety.^{12,39,41}

Marital status, presence of comorbidities, years in practice, receiving psychological support were also not statistically significant factors to mental health outcome in this study.

No variable have significant association with distress symptoms. Likewise the stepwise logistic regression (backward selection) has not identified any significant predictors of distress symptoms.

5. Conclusion and Recommendation

Healthcare workers are at risk not only from physical exhaustion, but also from emotional strain brought about by this pandemic. This will cause adverse outcomes not only for individual wellbeing, but also for patient care and the healthcare system.

Cognizant of high rates of depression, anxiety and distress among our younger health care workers living with relatives with comorbidities during this time will be a challenge to both hospital management and the government to establish and/or improve existing programs to promote mental wellbeing and address their mental health needs. These programs should be readily available and accessible to healthcare workers seeking psychological support. Psychological stress can accumulate over time and have an impact later in the outbreak, including posttraumatic stress disorder (PTSD), which should be investigated in future studies.

Appendix

APPENDIX I. Letter of Approval/Authorization to conduct study (Ancillary Division)

CARDINAL SANTOS MEDICAL CENTER
Department of Internal Medicine

June 16, 2020

Montserrat S. Velasquez, MD, DPBA
Chairman, Department of Anesthesiology and Pain Management
Head, Ancillary Division



Dear Dr. Velasquez,

Greetings!

As per requirements of our training program in the Department of Internal Medicine, I will be conducting a study entitled "Factors affecting mental health outcomes of health care workers during Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center". Data will be collected thru a written questionnaires or digital questionnaires thru google form which will be distributed to health care workers (physicians, nurses, nursing assistant, respiratory therapist, medical technologist, radiologic technologist, and cardiac therapist) with direct contact to patients with suspected or confirmed cases of COVID 19. We are respectfully requesting from your kind office that we be granted permission to carry out this study in this institution. Rest assured, we will maintain the confidentiality of the physician-patient relationship and will ensure the privacy of information.

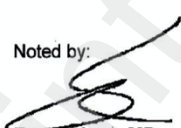
For further information, kindly refer to a copy of the Research Protocol that I have provided for your review.

Thank you for your kind consideration and assistance in this matter.

Sincerely yours,


Archimedes A. Apa-ap, MD
Third Year Resident
Department of Internal Medicine

Noted by:


Erwin Dizon, MD
Chairman
Department of Internal Medicine
Cardinal Santos Medical Center

Noted:

*pending approval of
the participants.*



APPENDIX II. Letter of Approval/Authorization to conduct study (Nursing Division)



CARDINAL SANTOS
MEDICAL CENTER
YOU are family

(02) 727 - 0001
10 Wilson St. Greenhills West
San Juan City, Metro Manila,
1502 Philippines



July 20, 2020

Candice Genuino-Montaño, MD
Head, Research Center
Cardinal Santos Medical Center

Dear Dr. Genuino-Montaño,

Good day!

This is to endorse Dr. Archimedes Apa-ap, 3rd year Internal Medicine Resident in Cardinal Santos Medical Center for research conduction.

His research protocol entitled "Factors Affecting Mental Health Outcomes of Healthcare Workers during Corona Virus Disease (COVID-19) Pandemic at Cardinal Santos Medical Center " was technically reviewed by the Nursing Training, Research & Quality Team of Cardinal Santos Medical Center.

Technical Reviewers were the following:

Ms. Leona Paula L. Macalintal RN, MAN (Director, Clinical Nursing Operations)

Mr. Julius T. Flores RN, MAN (Department Head of Nursing Training, Research and Quality Department)

Mr. Moises Edd O. Gangano RN, MAN (Team Leader of Nursing Research, Innovation and Patient Safety)

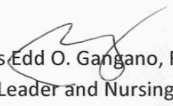
Purpose of the research protocol is to identify the factors affecting mental health outcomes of health care workers during Coronavirus Disease (COVID-19) pandemic at Cardinal Santos Medical Center.

Revisions recommended by the technical reviewers were already implemented in the said protocol, hence, being endorsed to CSMC Research Center for further review and endorsement to ethics board (RERC) for ethical approval prior to research conduction.

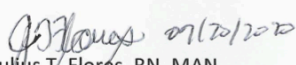
www.cardinalsantos.com.ph

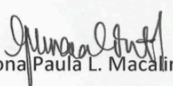


Best regards,


Moises Edd O. Gangano, RN, MAN
Team Leader and Nursing Research,
Innovation and Patient Safety

Noted by:


Julius T. Flores, RN, MAN
Department Head
Nursing Training, Research and Quality Department


Ms. Leona Paula L. Macalintal RN, MAN
Director
Clinical Nursing Operations

References

- ¹ Ren, L.-L.; Wang, Y.-M.; Wu, Z.-Q.; Xiang, Z.-C.; Guo, L.; Xu, T.; Jiang, Y.-Z.; Xiong, Y.; Li, Y.-J.; Li, X.-W.; et al. Identification of a novel coronavirus causing severe pneumonia in human: A descriptive study. *Chin. Med. J. (Engl.)* 2020. [CrossRef] [PubMed]
- ² World Health Organization (WHO). Mental Health and Psychosocial Considerations during the COVID-19 Outbreak. Available online: <https://www.who.int/docs/default-source/coronaviruse/mental-health-considerations.pdf> (accessed on 17 April 2020).
- ³ World Health Organization Coronavirus disease (COVID 2019) Situational Report September 27, 2020. Available online: <https://www.who.int/docs/default-source/coronaviruse/situation-reports/20201005-weekly-epi-update-8.pdf> (accessed on 7 October 2020)
- ⁴ Department of Health COVID 19 tracker Republic of the Philippines Available online: <https://www.doh.gov.ph/covid19tracker> (accessed on 7 October 2020)
- ⁵ Department of Health Republic of The Philippines Available online: <https://www.rappler.com/nation/health-workers-coronavirus-cases-philippines-august-3-2020> (accessed on 7 October 2020)
- ⁶ Gouliou, P.; Mantas, C.; Dimitroula, D.; Mantis, D.; Hyphantis, T. General hospital staff worries, perceived sufficiency of information and associated psychological distress during the A/H1N1 influenza pandemic. *BMC Infect. Dis.* 2010, 10, 322.
- ⁷ Chan, A.O.M.; Huak, C.Y. Psychological impact of the 2003 severe acute respiratory syndrome outbreak on health care workers in a medium size regional general hospital in Singapore. *Occup. Med.* 2004, 54, 190–196.
- ⁸ McAlonan, G.M.; Lee, A.M.; Cheung, V.; Cheung, C.; Tsang, K.W.T.; Sham, P.C.; Chua, S.E.; Wong, J.G.W.S. Immediate and Sustained Psychological Impact of an Emerging Infectious Disease Outbreak on Health Care Workers. *Can. J. Psychiatry* 2007, 52, 241–247.
- ⁹ Ayanian, J.Z. Editor 's Comment: Mental Health Needs of Health Care Workers Providing Frontline COVID-19 Care. Available online: <https://jamanetwork.com/channels/health-forum/fullarticle/2764228> (accessed on 17 April 2020).
- ¹⁰ Hogan, L. *The 'Psychological First Aid' Helping Healthcare Workers through Crisis*; RTE Ireland's National Television and Radio Broadcaster, 2020; Available online: <https://www.rte.ie/news/coronavirus/2020/0409/1129401-covid19-coronavirus-mental-health-medical-staff/> (accessed on 17 April 2020).
- ¹¹ Patel, R.S.; Bachu, R.; Adikay, A.; Malik, M.; Shah, M. Factors Related to Physician Burnout and Its Consequences: A Review. *Behav. Sci.* 2018, 8, 98.
- ¹² Lai, J. Et. Al., Factors Associated With Mental Health Outcomes Among Health Care Workers Exposed to Coronavirus Disease 2019. *JAMA network* 2020 doi:10.1001/jamanetworkopen.2020.3976
- ¹³ Kang, L. Et. Al., Impact on mental health and perceptions of psychological care among medical and nursing staff in Wuhan during the 2019 novel coronavirus disease outbreak: A cross-sectional study. *Brain Behavior and Immunity* 2020. <https://doi.org/10.1016/j.bbi.2020.03.028>
- ¹⁴ McEachan R, Taylor N, Harrison R, Lawton R, Gardner P, Conner M. Meta-analysis of the reasoned action approach (RAA) to understanding health behaviors. *Ann Behav Med* 2016;50: 592e612.
- ¹⁵ COVID 19 and Diabetes. International Federation of Diabetes. Available from: <https://www.idf.org/aboutdiabetes/what-is-diabetes/covid-19-and-diabetes/1-covid-19-and-diabetes.html> [Accessed on 30th May 2020].
- ¹⁶ Wu Z, McGoogan JM. Characteristics of and important lessons from the coronavirus disease 2019 (COVID-19) outbreak in China: summary of a report of 72 314 cases from the Chinese center for disease control and prevention. *J Am Med Assoc* 2020 Feb 24. <https://doi.org/10.1001/jama.2020.2648>.

- ¹⁷ Zhang, M., Et. Al., Knowledge, attitude, and practice regarding COVID-19 among healthcare workers in Henan, China. *Journal of Hospital Infection* 2020. <https://doi.org/10.1016/j.jhin.2020.04.012>
- ¹⁸ Lipley, N. Covid-19: Not a 'Mental Health Crisis', Healthcare Experts Warn. Available online: <https://rcni.com/nursing-standard/newsroom/news/COVID-19-not-a-mental-health-crisis-healthcare-experts-warn-159611> (accessed on 17 April 2020).
- ¹⁹ Ayanian, J.Z. Editor 's Comment: Mental Health Needs of Health Care Workers Providing Frontline COVID-19 Care. Available online: <https://jamanetwork.com/channels/health-forum/fullarticle/2764228> (accessed on 17 April 2020).
- ²⁰ Shyrock, T. COVID-19 Raises Ethical Dilemmas for Many Physicians. Available online: <https://www.medicaleconomics.com/news/covid-19-raises-ethical-dilemmas-many-physicians> (accessed on 17 April 2020).
- ²¹ Maunder R, Hunter J, Vincent L, et al. The immediate psychological and occupational impact of the 2003 SARS outbreak in a teaching hospital. *CMAJ*. 2003;168(10):1245-1251.
- ²² Centers for Disease Control and Prevention. Mental Health. Available online: <https://www.cdc.gov/mentalhealth/index.htm> (accessed on 24 May 2020)
- ²³ American Psychiatric Association. Available online: <https://www.psychiatry.org/patients-families/depression/what-is-depression> (accessed on 15 October 2020).
- ²⁴ Kocalevent, R.D., et al., 2013. Standardization of the depression screener patient health questionnaire (PHQ-9) in the general population. *Gen. Hospital Psychiat.* 35, 551–555. <https://doi.org/10.1016/j.genhosppsych.2013.04.006>.
- ²⁵ American Psychological Association. Available online: <https://www.apa.org/search?query=anxiety> (accessed on 15 October 2020).
- ²⁶ Löwe, B., et al., 2008. Validation and standardization of the generalized anxiety disorder screener (GAD-7) in the general population. *Med. Care* 46, 266–274. <https://doi.org/10.1097/mlr.0b013e318160d093>.
- ²⁷ Mental Health and Wellness. Available online: <https://brocku.ca/mental-health/signs-of-distress/#physical> (accessed on 15 October 2020).
- ²⁸ Daniel S., Weiss, Ph.D., 2007. *The Impact of Event Scale: Revised*. Guilford Press, New York, pp. 399–411.
- ²⁹ WHO, 2020. Statement on the second meeting of the International Health Regulations (2005) Emergency Committee regarding the outbreak of novel coronavirus (2019- nCoV). [https://www.who.int/news-room/detail/30-01-2020-statement-on-the-second-meeting-of-the-international-health-regulations-\(2005\)-emergency-committee-regarding-the-outbreak-of-novel-coronavirus-\(2019-ncov\)](https://www.who.int/news-room/detail/30-01-2020-statement-on-the-second-meeting-of-the-international-health-regulations-(2005)-emergency-committee-regarding-the-outbreak-of-novel-coronavirus-(2019-ncov)) (assessed Feb 15, 2020).
- ³⁰ WHO. Q&A on coronaviruses (COVID-19). 2020. [Online]. Available from: <https://www.who.int/news-room/q-a-detail/q-a-coronaviruses> [Accessed on 1st March 2020].
- ³¹ Gao, H. et. Al., Knowledge and attitude toward COVID-19 among healthcare workers at District 2 Hospital, Ho Chi Minh City. *Asia Pacific Journal of Tropical Medicine*. April 2020. <https://www.researchgate.net/publication/340413844>
- ³² Olum, R., et. Al., Knowledge, Attitude and Practices of health care workers at Makerere University Teaching Hospitals, Uganda. *Frontiers of Public Health*. April 30, 2020.
- Ball, G.H., 1967. A clustering technique for summarizing multivariate data. *Behav. Sci.* <https://doi.org/10.1002/bs.3830120210>.
- ³³ <https://www.cdc.gov/coronavirus/2019-ncov/hcp/using-ppe.html> (Accessed July 10, 2020)
- ³⁴ https://www.cdc.gov/coronavirus/2019-ncov/downloads/A_FS_HCP_COVID19_PPE_11x17.pdf (Accessed July 10, 2020)
- ³⁵ Ahmed, M.Z., Ahmed, O., Zhou, A., Sang, H., Liu, S., Ahmad, A., 2020. Epidemic of COVID-19 in China and associated psychological problems. *Asian J. Psychiatr.* 51, 102092. <https://doi.org/10.1016/j.ajp.2020.102092>.

- ³⁶ Gao, J., Zheng, P., Jia, Y., Chen, H., Mao, Y., Chen, S., Wang, Y., Fu, H., Dai, J., 2020. Mental health problems and social media exposure during COVID-19 outbreak. PLoS ONE 15 (4), e0231924. <https://doi.org/10.1371/journal.pone.0231924>.
- ³⁷ Huang, Y., Zhao, N., 2020. Generalized anxiety disorder, depressive symptoms and sleep quality during COVID-19 outbreak in China: a web-based cross-sectional survey. Psychiatry Res 288, 112954. <https://doi.org/10.1016/j.psychres.2020.112954>.
- ³⁸ Cao, W., Fang, Z., Hou, G., Han, M., Xu, X., Dong, J., Zheng, J., 2020. The psychological impact of the COVID-19 epidemic on college students in China. Psychiatry Res 287, 112934. <https://doi.org/10.1016/j.psychres.2020.112934>.
- ³⁹ Cai, H., Tu, B., Ma, J., et al., 2020. Psychological impact and coping strategies of frontline medical staff in Hunan between January and March 2020 during the outbreak of Coronavirus Disease 2019 (COVID19) in Hubei, China. Med. Sci. Monitor 26, e924171. <https://doi.org/10.12659/MSM.924171>.
- ⁴⁰ Louie, P.K., Harada, G.K., McCarthy, M.H., et al., 2020. The Impact of COVID-19 Pandemic on Spine Surgeons Worldwide. Glob. Spine J. <https://doi.org/10.1177/2192568220925783>.
- ⁴¹ Li, G., Miao, J., Wang, H., et al., 2020a. Psychological impact on women health workers involved in COVID-19 outbreak in Wuhan: a cross-sectional study. J. Neurol. Neurosurg. Psychiatr. <https://doi.org/10.1136/jnnp-2020-323134>.
- ⁴² Amin, F., et.al., COVID-19 Pandemic-Knowledge, Perception, Anxiety and Depression among frontline doctors of Pakistan. Research square. <https://doi.org/10.21203/rs.3.rs-27559/v1>